

Investigating coherence of Atlantic Meridional Heat Transport

Patterns of propagation and their drivers

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Master thesis objectives

- 1st objective:

How coherent is MHT over the Atlantic ?

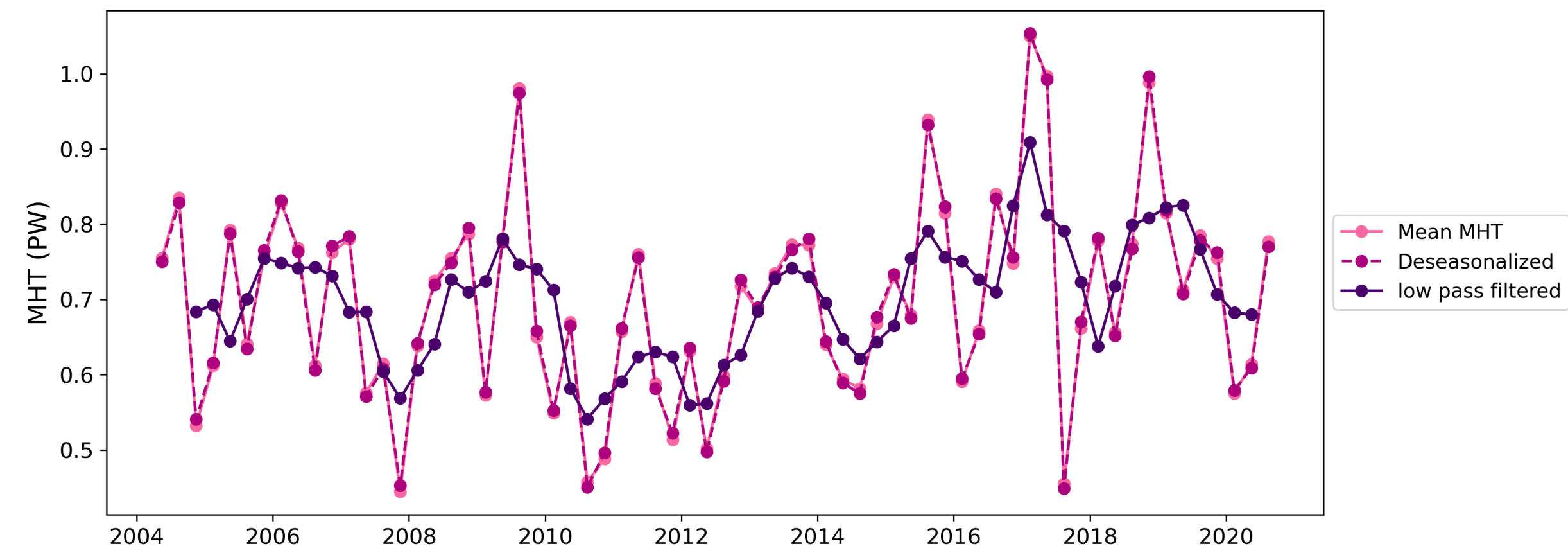
- 2nd objective:

What drives the incoherence? Surface Heat Flux vs. Ocean Circulation change

Data:

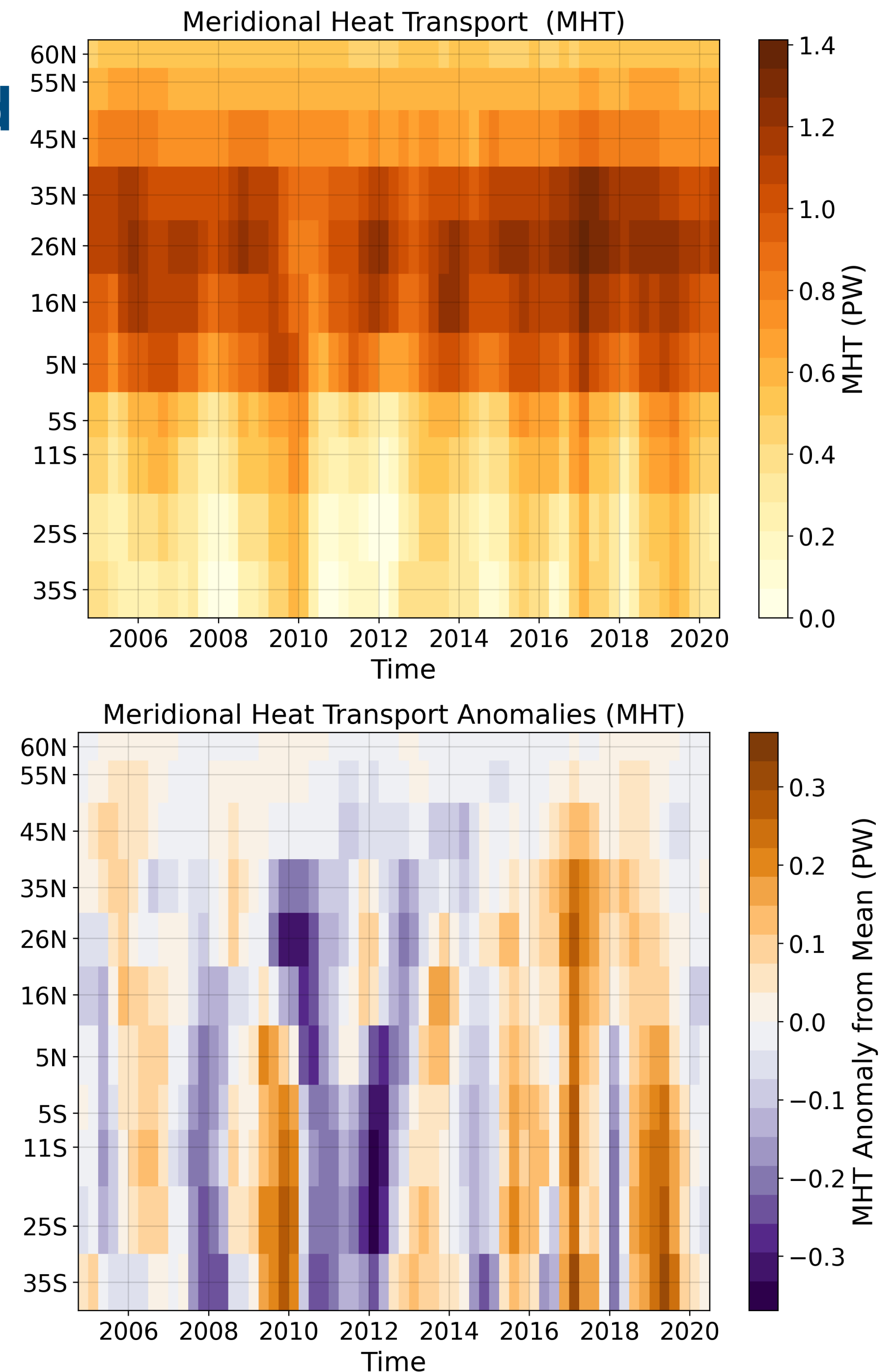
Bayesian estimates of MHT from Calafat et al. (2025).

- Deseasonalized
 - 1-yr low pass filtered (running-mean)
- interannual variability



MHT spatial and temporal structure hints toward meridional coherent anomaly bands

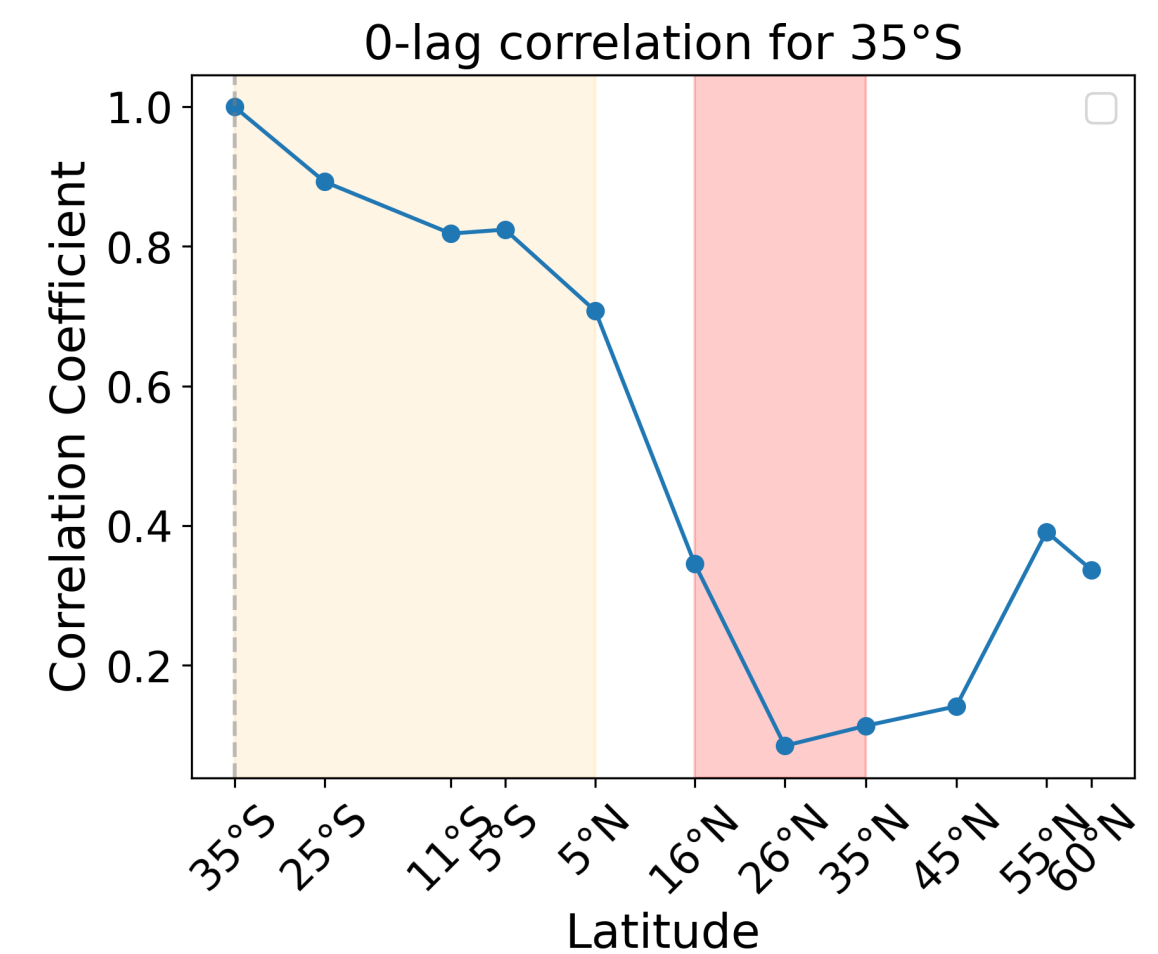
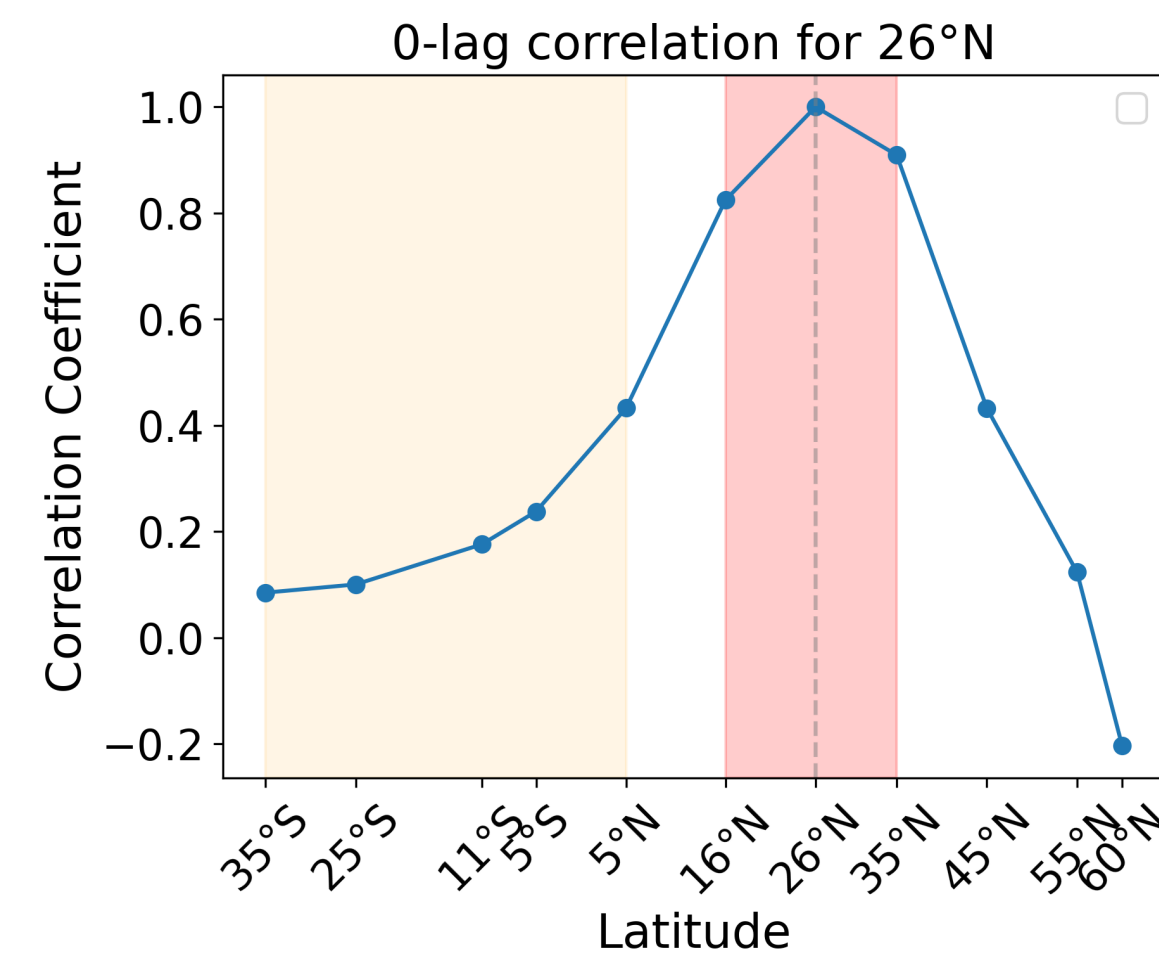
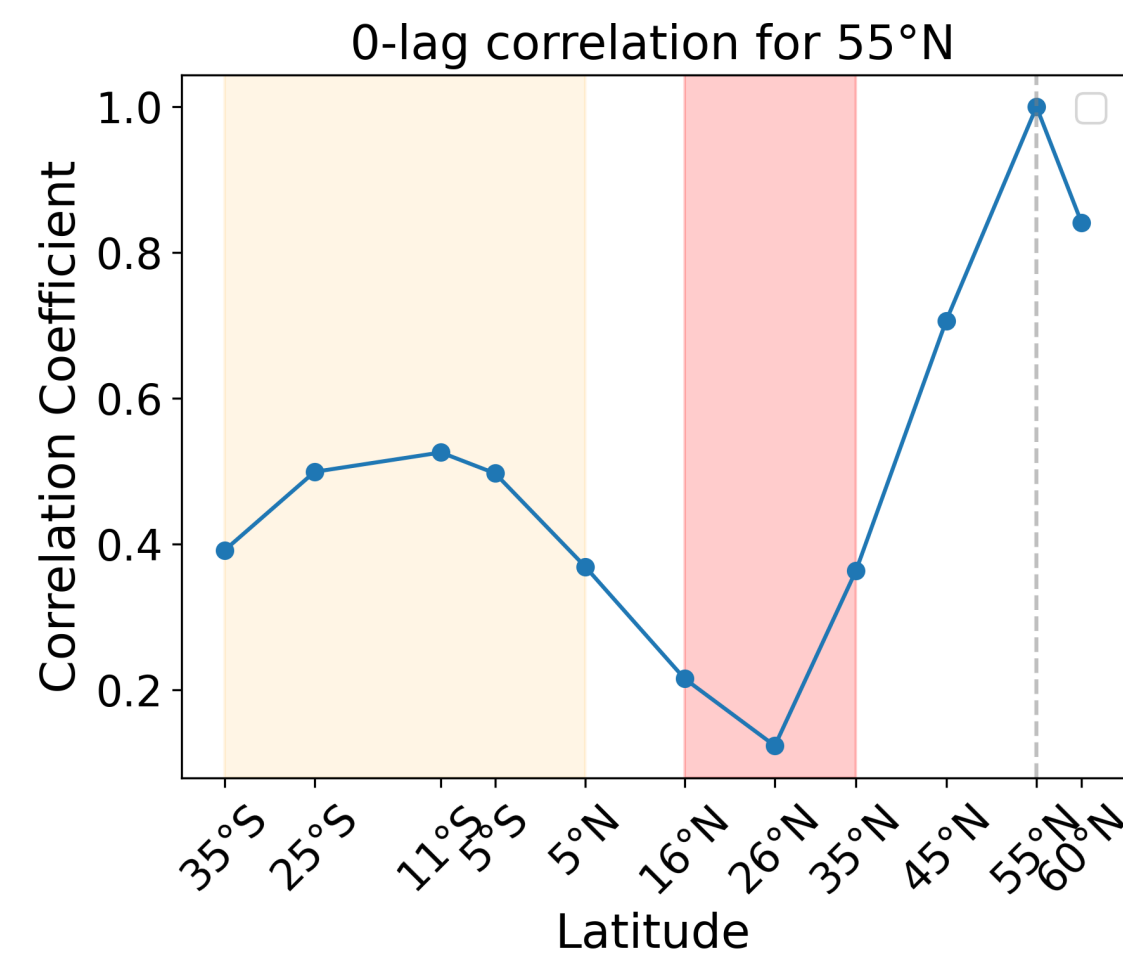
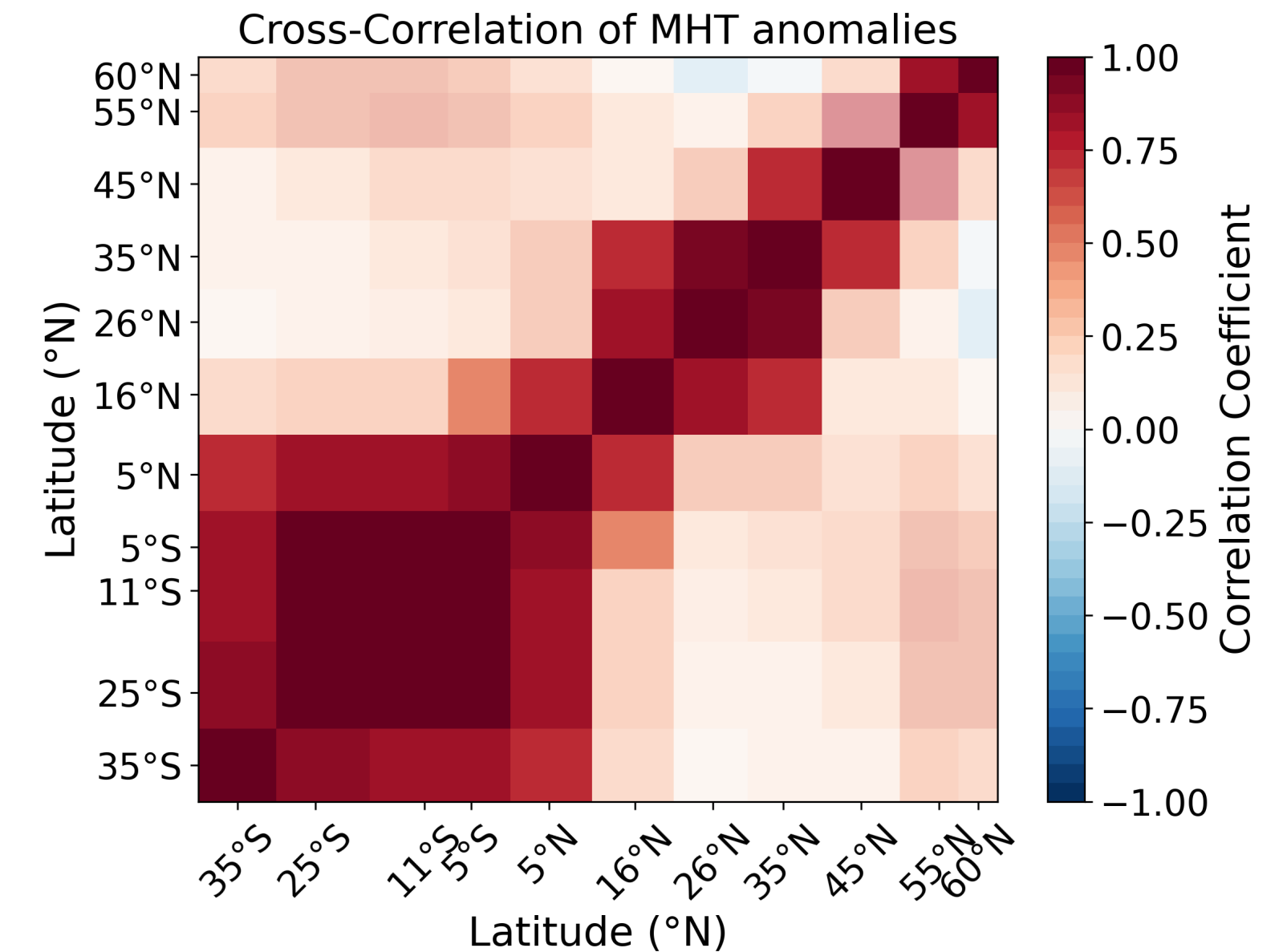
- Anomalies are computed for each latitude band's individual time mean
- Strongest MHT over tropics, maximum at 26°N
- Smallest anomalies at northern latitudes - likely reflecting the MHT=0 constraint at 65°N
- Basin wide pattern for specific years: 2006, 2008, 2010, 2011, 2015, 2017, 2019
- next: 0-lag and cross-correlation confirm covarying bands, followed by lead/lag analysis to see anomaly propagation



MHT - cross correlation (0 lag)

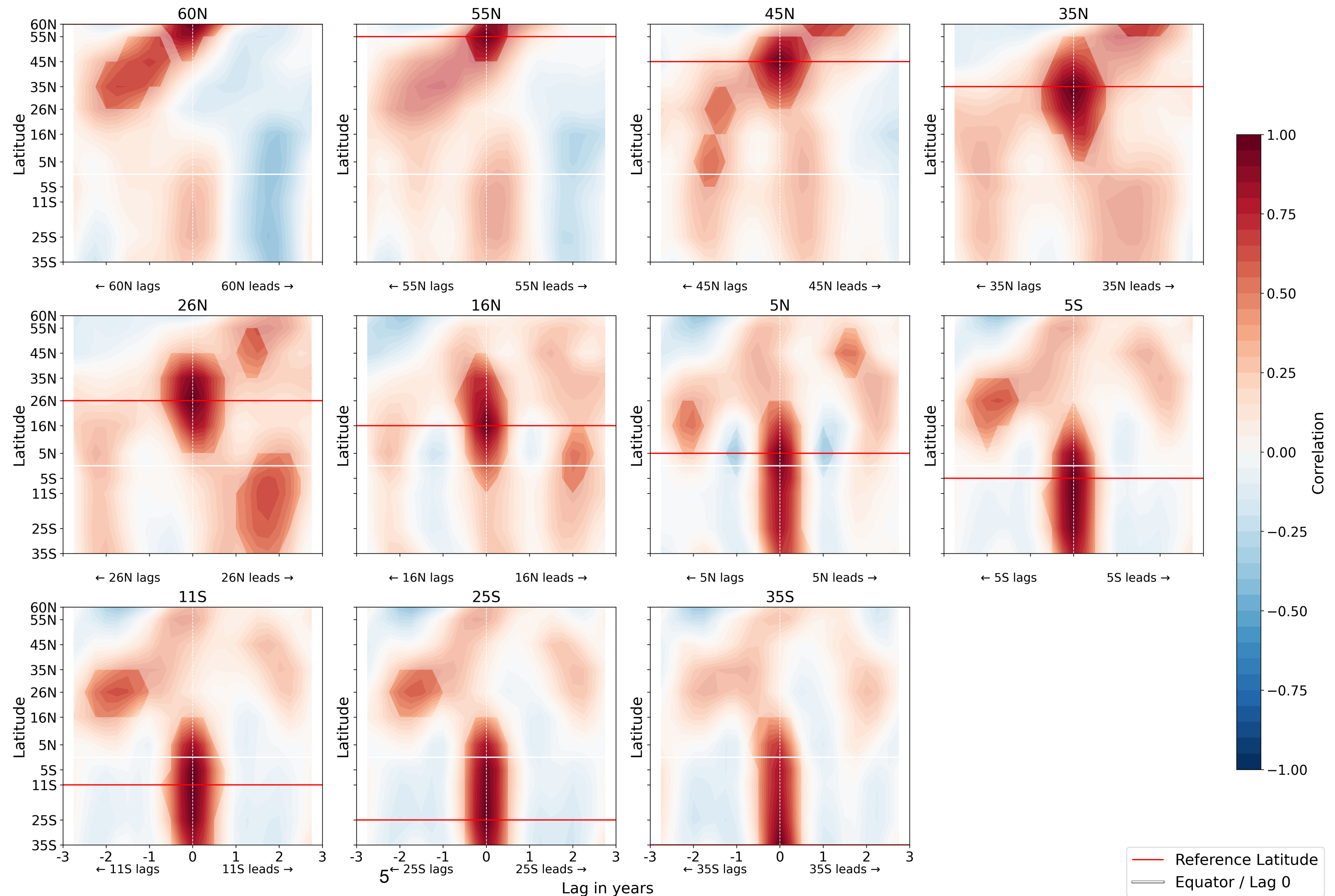
Calafat et al. (2025): „*The MHT variability is not coherent across the whole Atlantic Ocean but instead covaries within, although not between, two distinct bands of latitudes, namely between 35 and 16°N and from 5°N to 35°S.*“

- Cross correlation (white shaded mask for significance) shows the 2 bands as significant
- 0-lag correlation confirms 2 bands:
 - 35 - 16°N (red) and 5°N - 35°S (yellow)
- Potentially third band at North Atlantic high latitudes (45 - 60°N), but not significant



Lead/lag with significance mask

- Significance mask (white) by calculating integral time scale and N_{eff} for critical correlation for each latitude
- Lag in years relative to reference latitude
- North Atlantic high latitudes show 1-2 years lag compared to North Atlantic lower latitudes
- South Atlantic seems overall coherent with a possible 2 year lag to 26°N



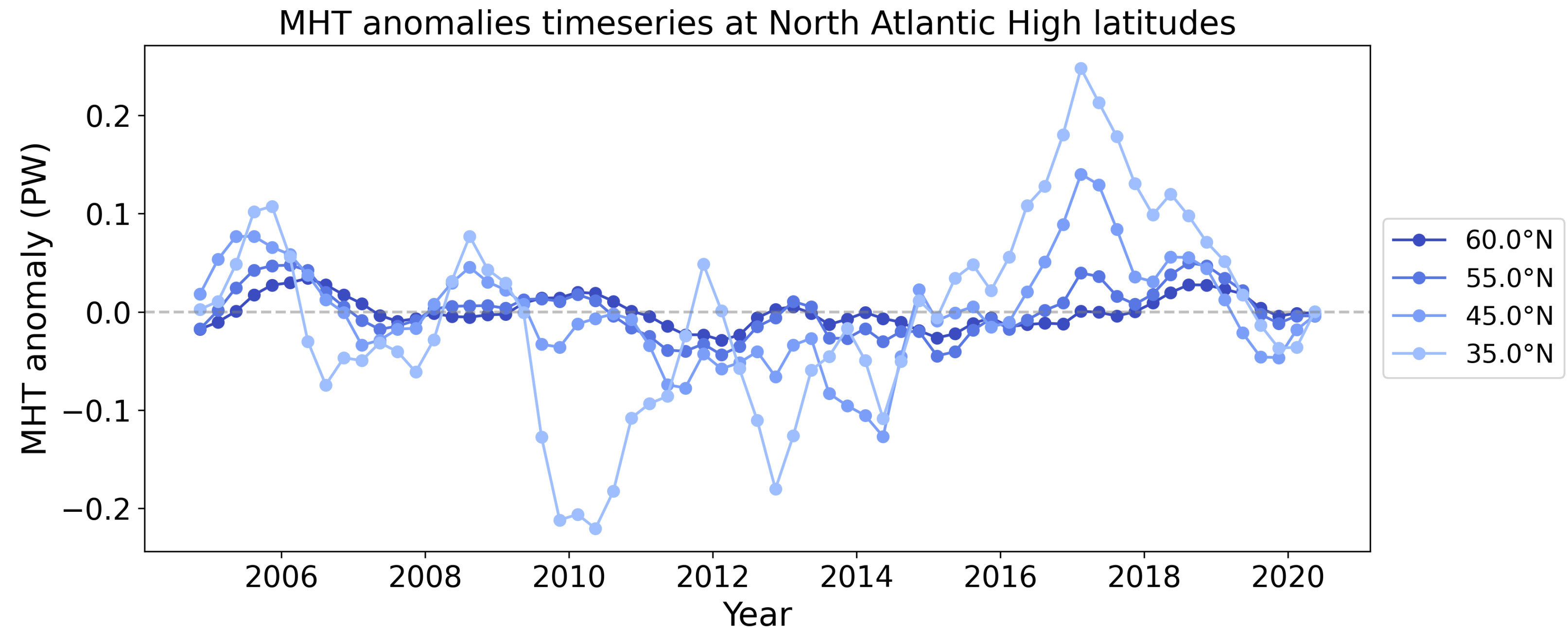
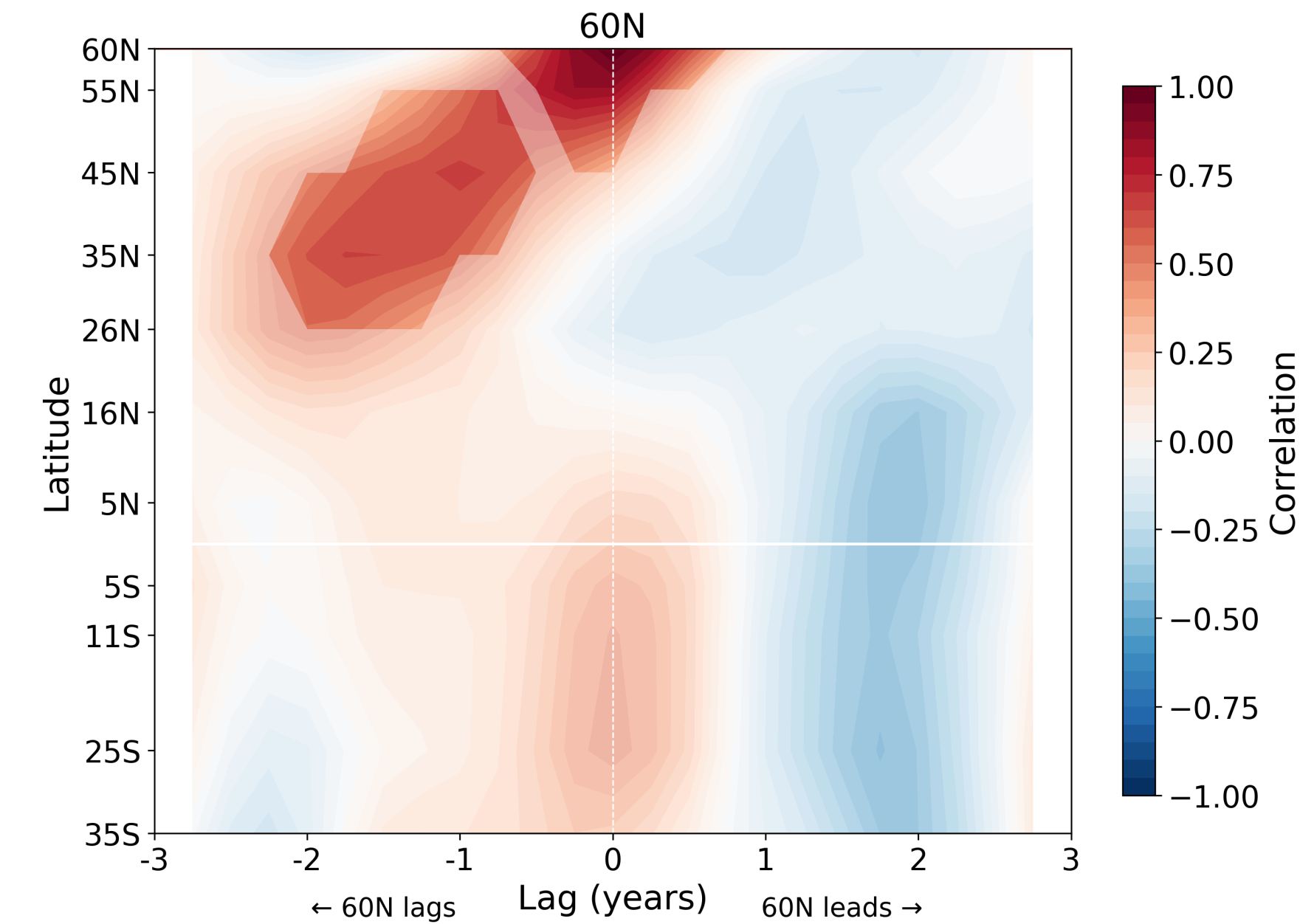
Do we see the lead/lag in the anomaly time series? (Work in progress)

Regions: North Atlantic High, North Atlantic Low, Equatorial and South Atlantic

- Check whether the MHT anomaly time series shows the same pattern indicated by the lead/lag analysis
- For each region find best shift, relative to northern reference latitude with choosing highest correlation and lowest RMSD and STD
- Compare shifts with timescales suggested by lead/lag plots

60°N compared with 45°N and 35°N

North Atlantic high latitudes



60-55°N lags:

45°N by 1-1.5 yrs

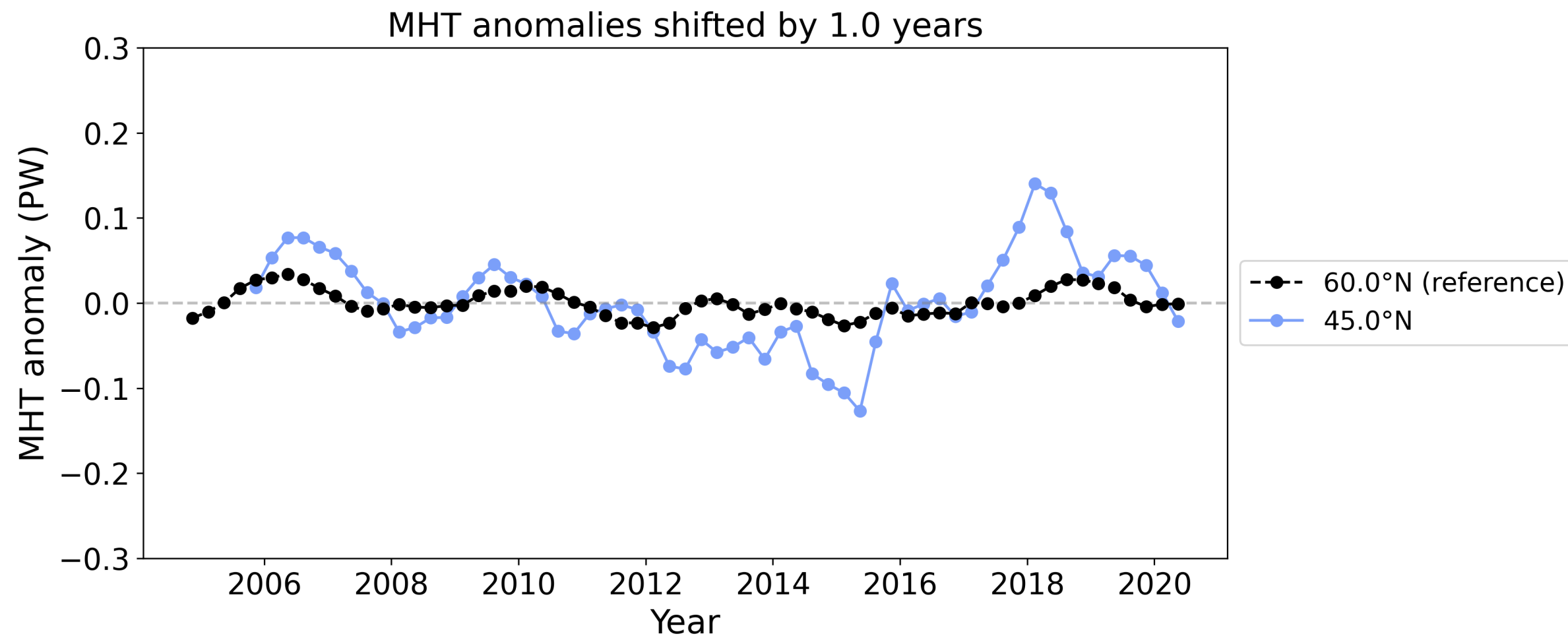
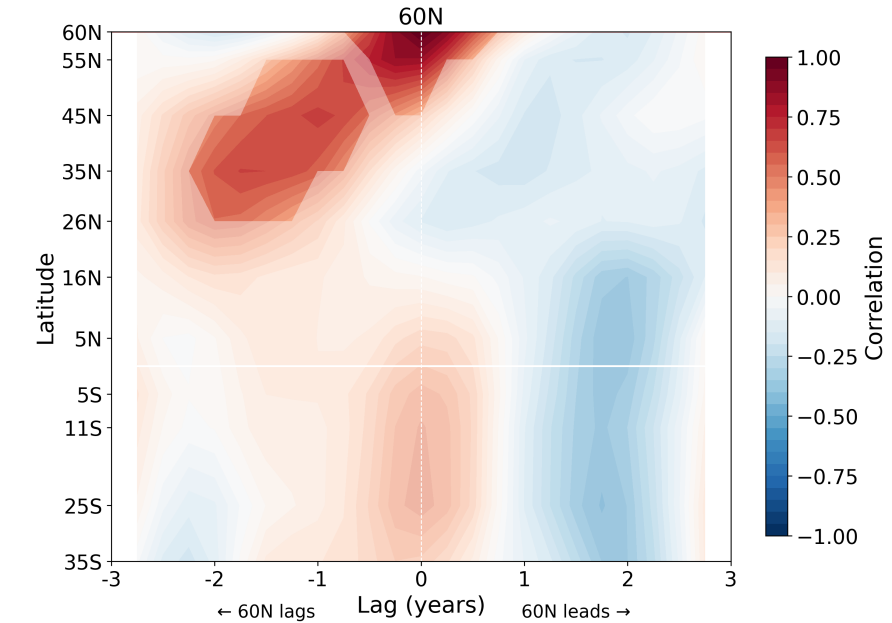
35°N by 1.5-2 yrs

From time series alone:

- No clear lead/lag pattern visible
- Coherent pattern at specific time periods

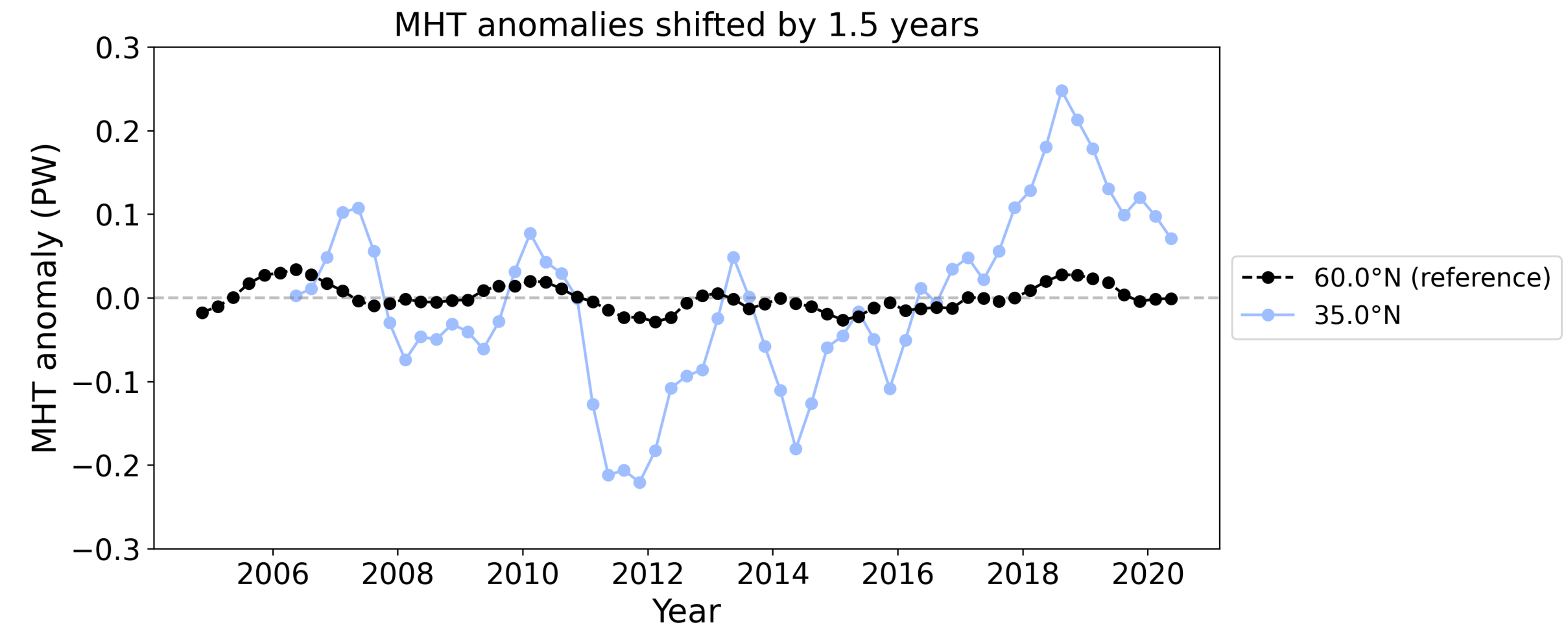
Applying the best fit shift

North Atlantic high latitudes - shifted relative to 60°N



60°N - 45°N best shift:

- 1 yr, corr = 0.65 (RMSE = 0.04, STD = 0.05)
- 60°N lags 45°N by 1 year
- 60°N stays weak while 45°N anomalies grow
- timescale (1-1.5yrs) consistent with lead/lag plot

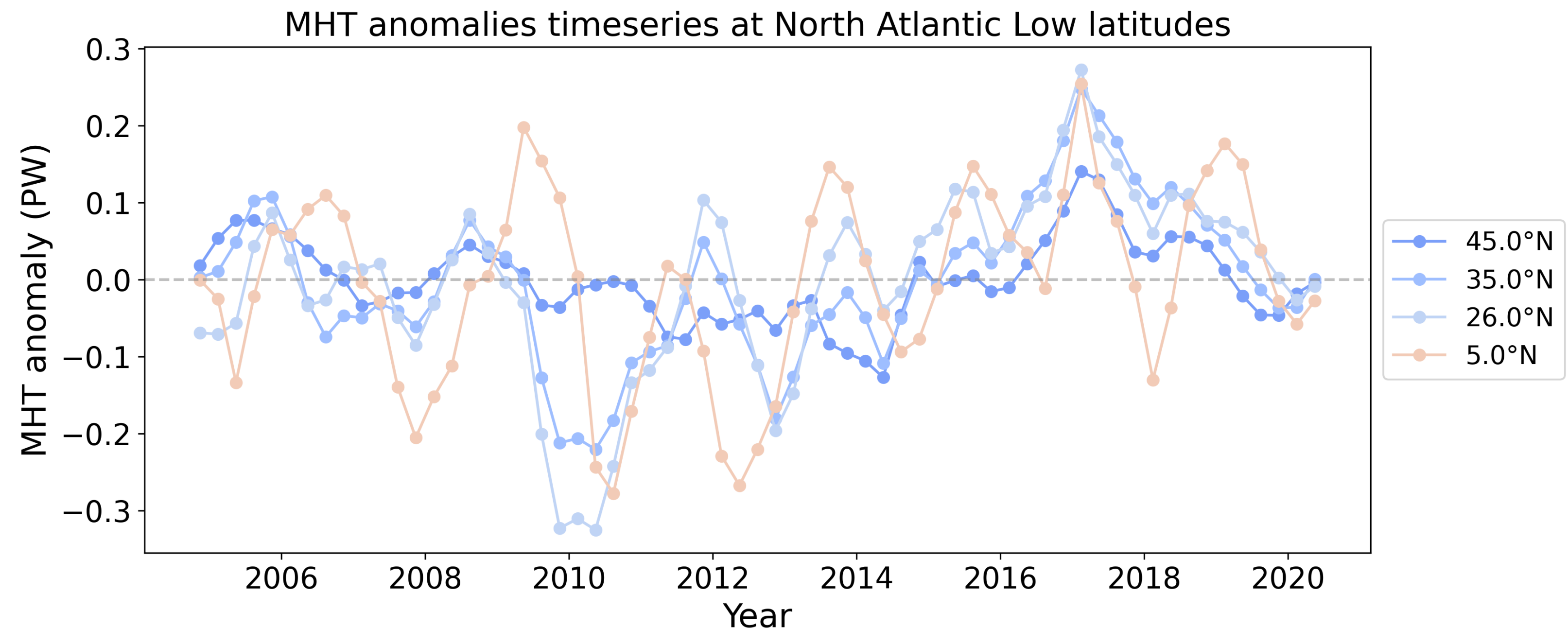
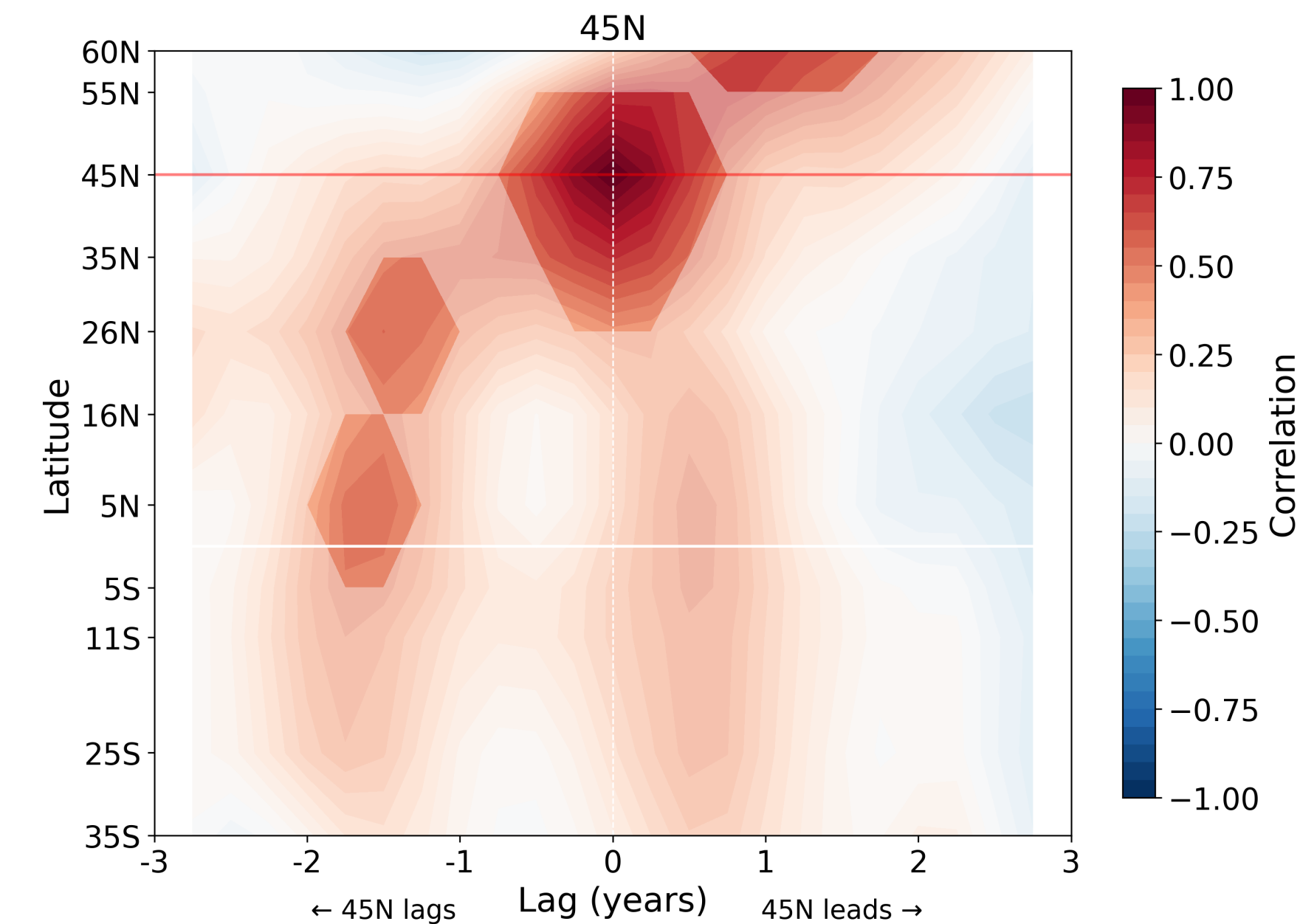


60°N - 35°N best shift:

- 1.5 yrs, corr = 0.65 (RMSE = 0.09, STD = 0.10)
- 60°N lags 35°N by 1.5 years - consistent timing as well
- Large RMSE/STD reflects a high amplitude mismatch between the two latitudes
- Open question: does high correlation really represent true lead/lag ?

45°N compared with 26°N and 5°N

North Atlantic low latitudes



45°N lags:

26°N by 1.5 yrs

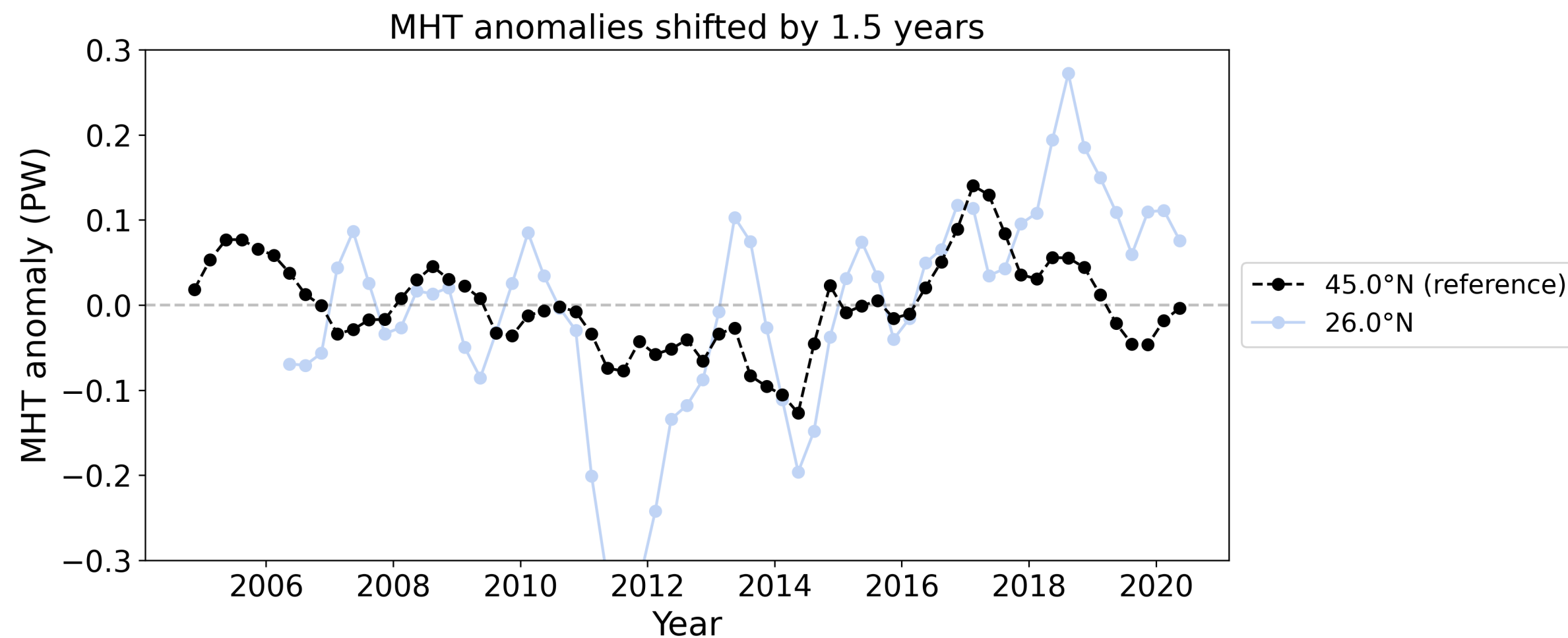
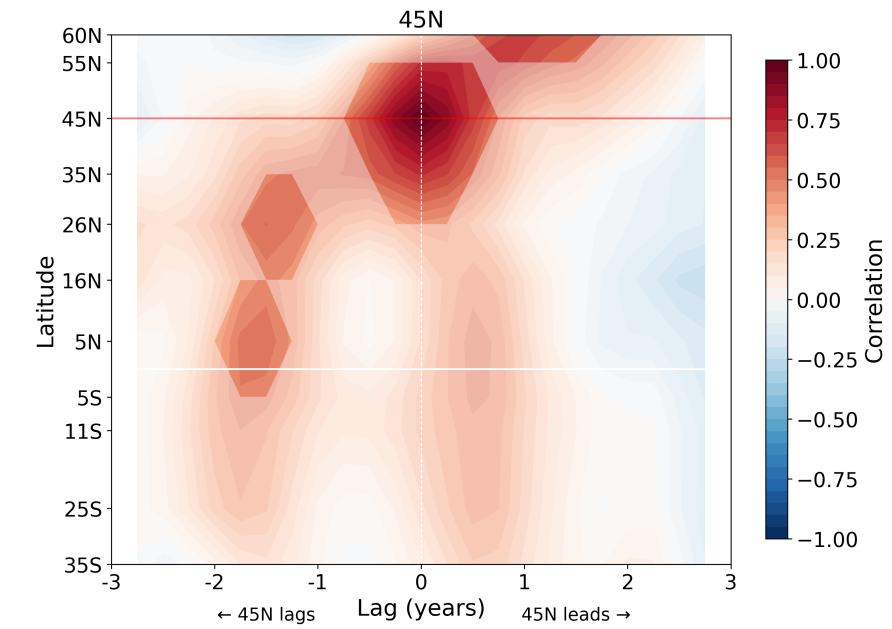
5°N by 1.5-2 yrs

From time series alone:

- No overall coherent or lead/lag pattern visible
- 45-26°N in phase at specific time periods (2005-2009, 2014-2020) while 5°N looks out of phase

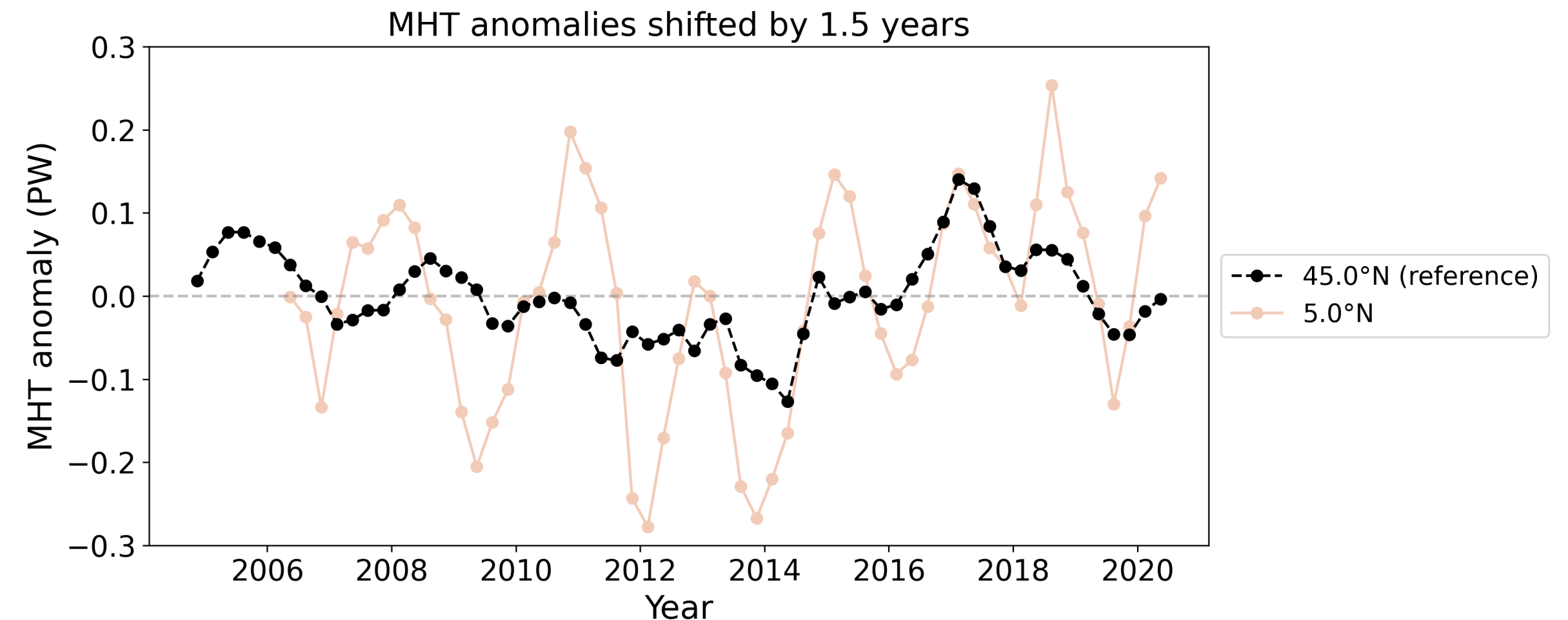
Applying the best-fit shift

North Atlantic high latitudes - shifted relative to 45°N



45°N - 26°N best shift :

- 1.5 yrs, corr = 0.54 (RMSE = 0.10, STD = 0.11)



45°N - 5°N best shift:

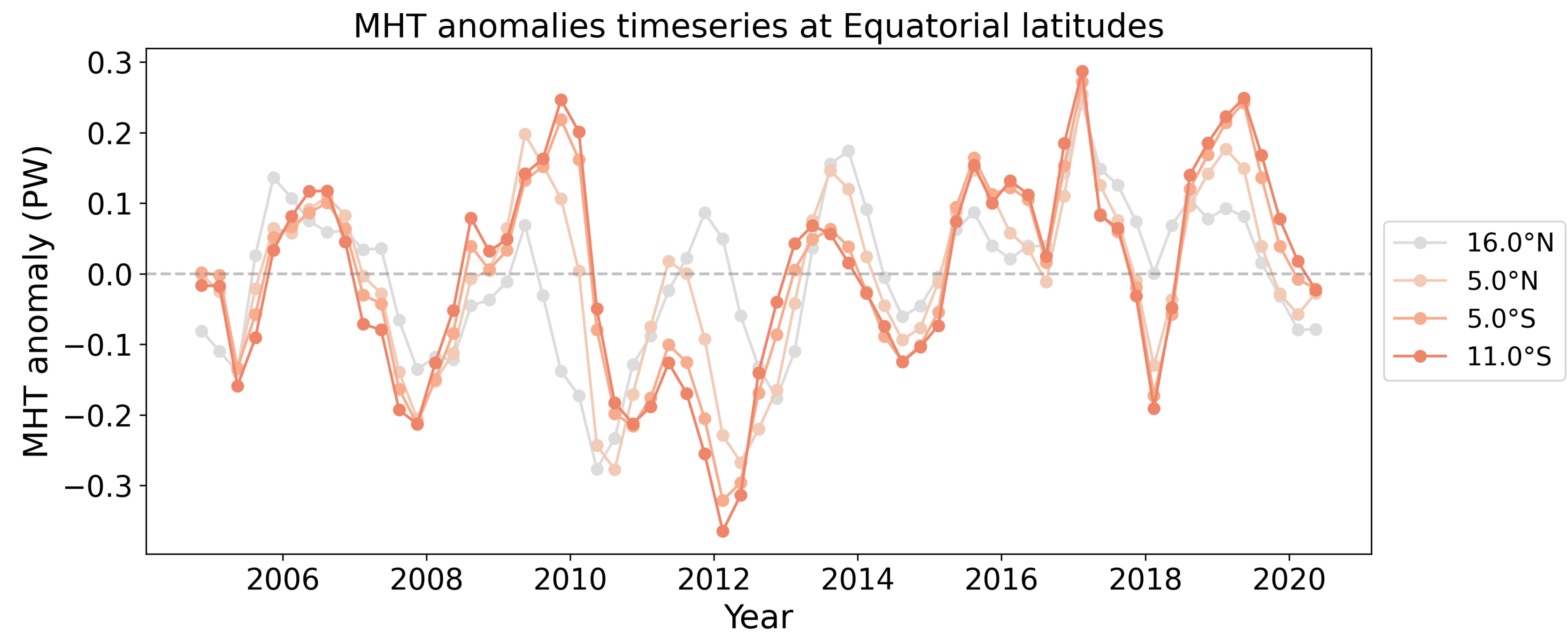
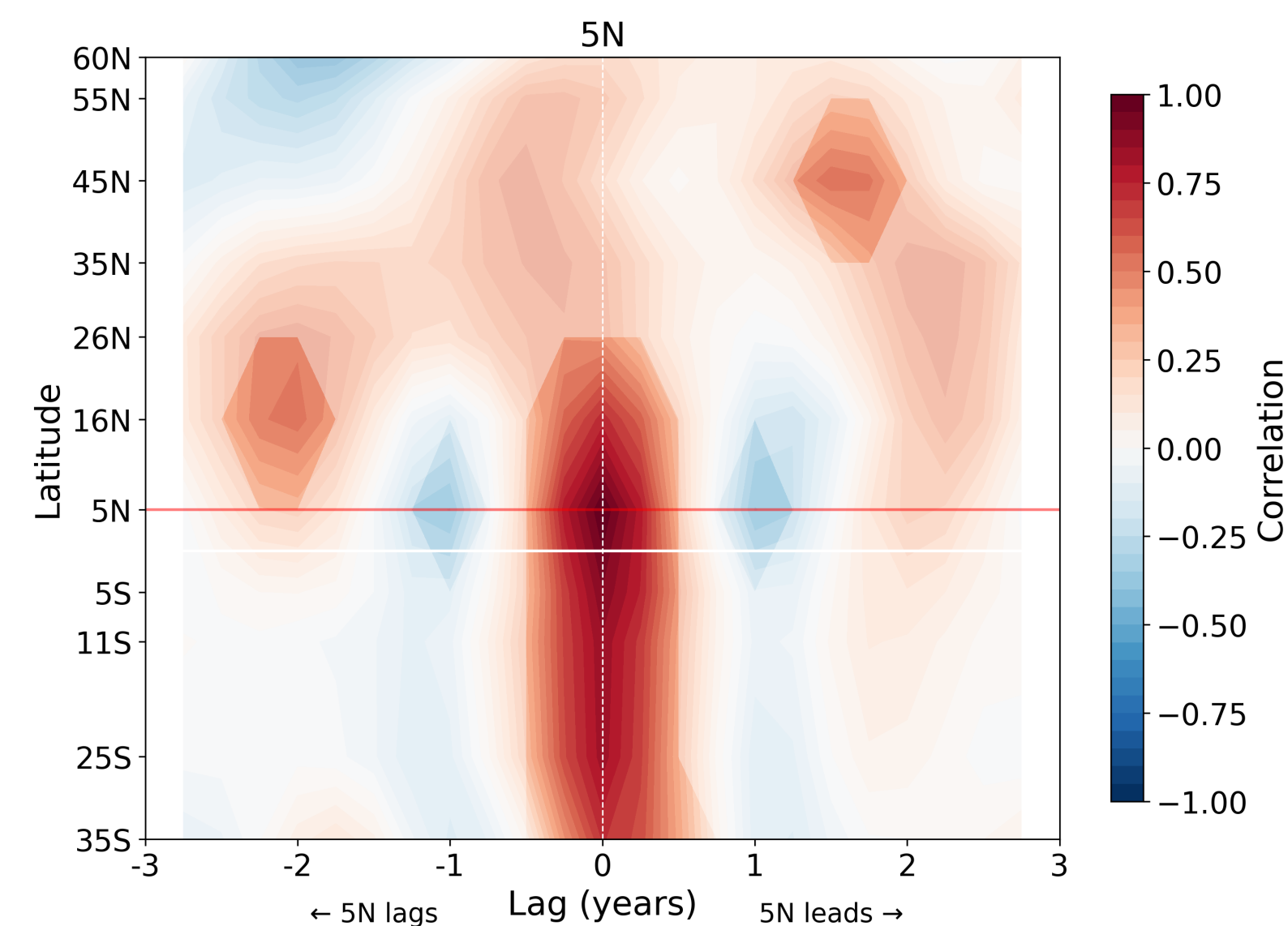
- 1.5 yrs, corr = 0.49 (RMSE = 0.11, STD = 0.12)

Both:

- Well matched from 2014-2018, here 45°N lags 26°N and 5°N by 1.5 yrs - timescale consistent with lead/lag plot
- Before 2014 out of phase
- Overall strong amplitude mismatches, more pronounced for 5°N

Equatorial latitudes: lead/lag= 0 — time series in phase

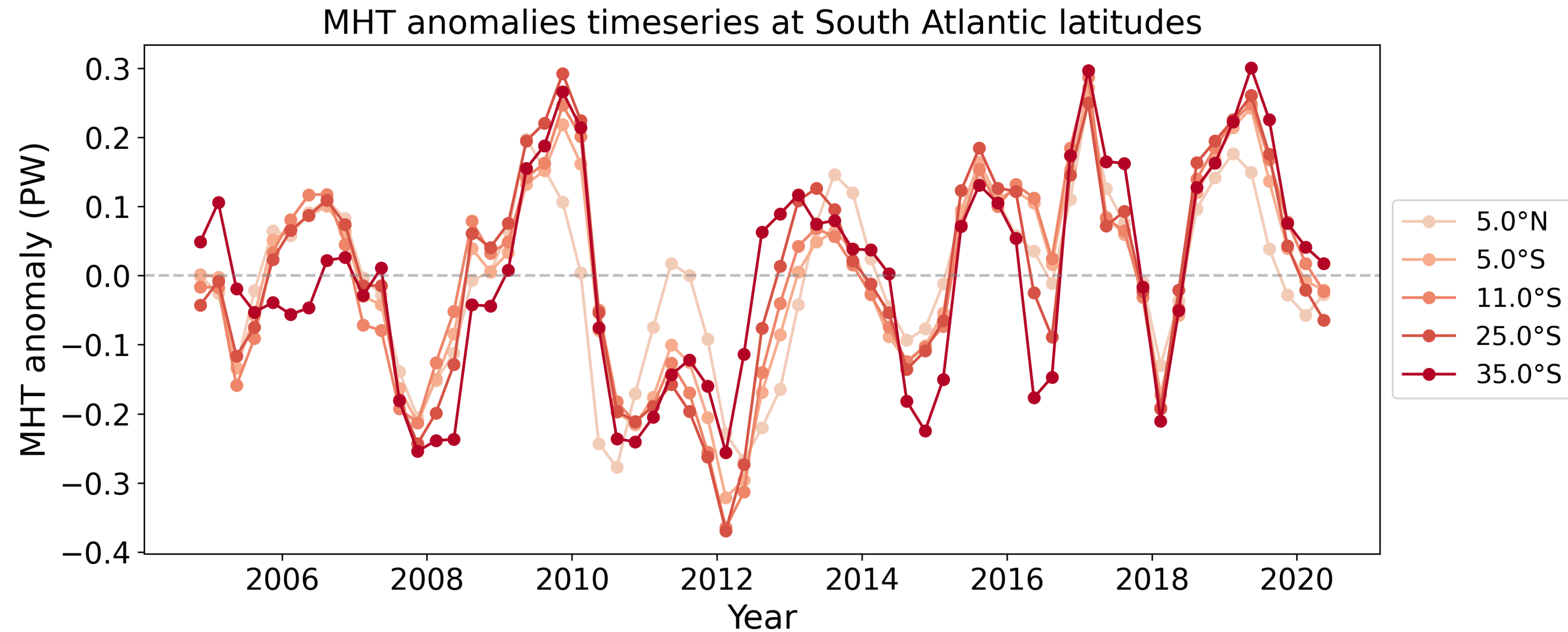
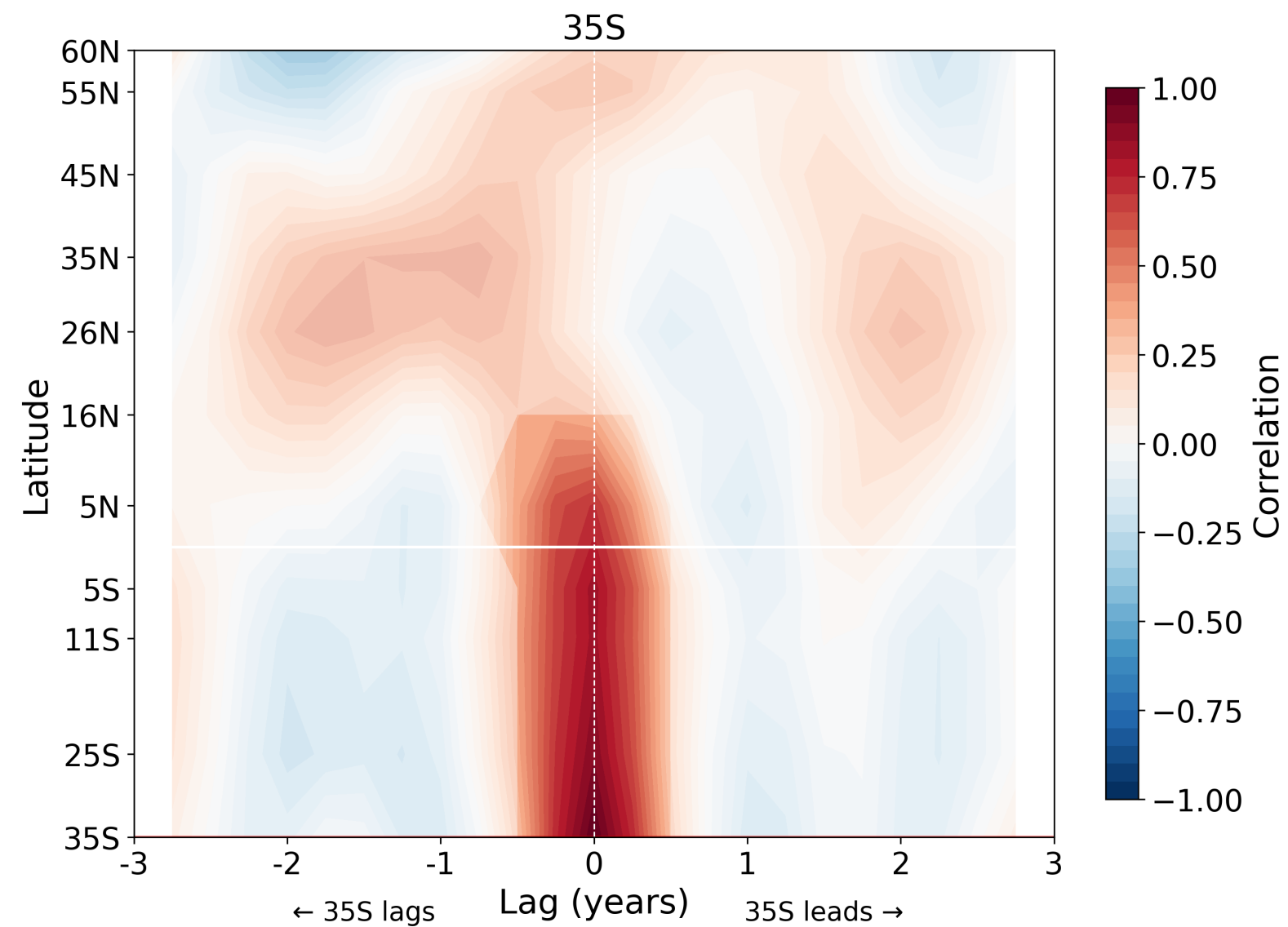
16°, 5°N, 5° and 11°S



Equatorial region **overall coherent**

South Atlantic: lead/lag= 0 — time series in phase

Including 5°N



South Atlantic (including 5°N) latitudes **overall coherent**

First conclusions for confirming lead/lag analysis

- **Timescale** from best-shift **matches** with those found in lead/lag analysis
- However overall **amplitude mismatches**, especially from 2010-2014
- Some regional and temporal matches include:
 - 60°N lags by 1.5, for 2005-2011 phase and amplitude matches
 - 45, 26 and 5°N mostly **out of phase until 2014**, then amplitude and phase well matched for 1.5 yrs shift
 - 45 and 26°N coherent until 2009 - large drop (likely due to AMOC drop due to record low NAO)
- High correlation at best shift doesn't show same amplitude, RMSD often comparable to the reference signal's full range - caution interpreting lead/lag
- Overall:
 - Consistent across pairs: **northern latitudes lag more equatorward ones** → propagation south to north
 - Equatorial region and South Atlantic **overall coherent**

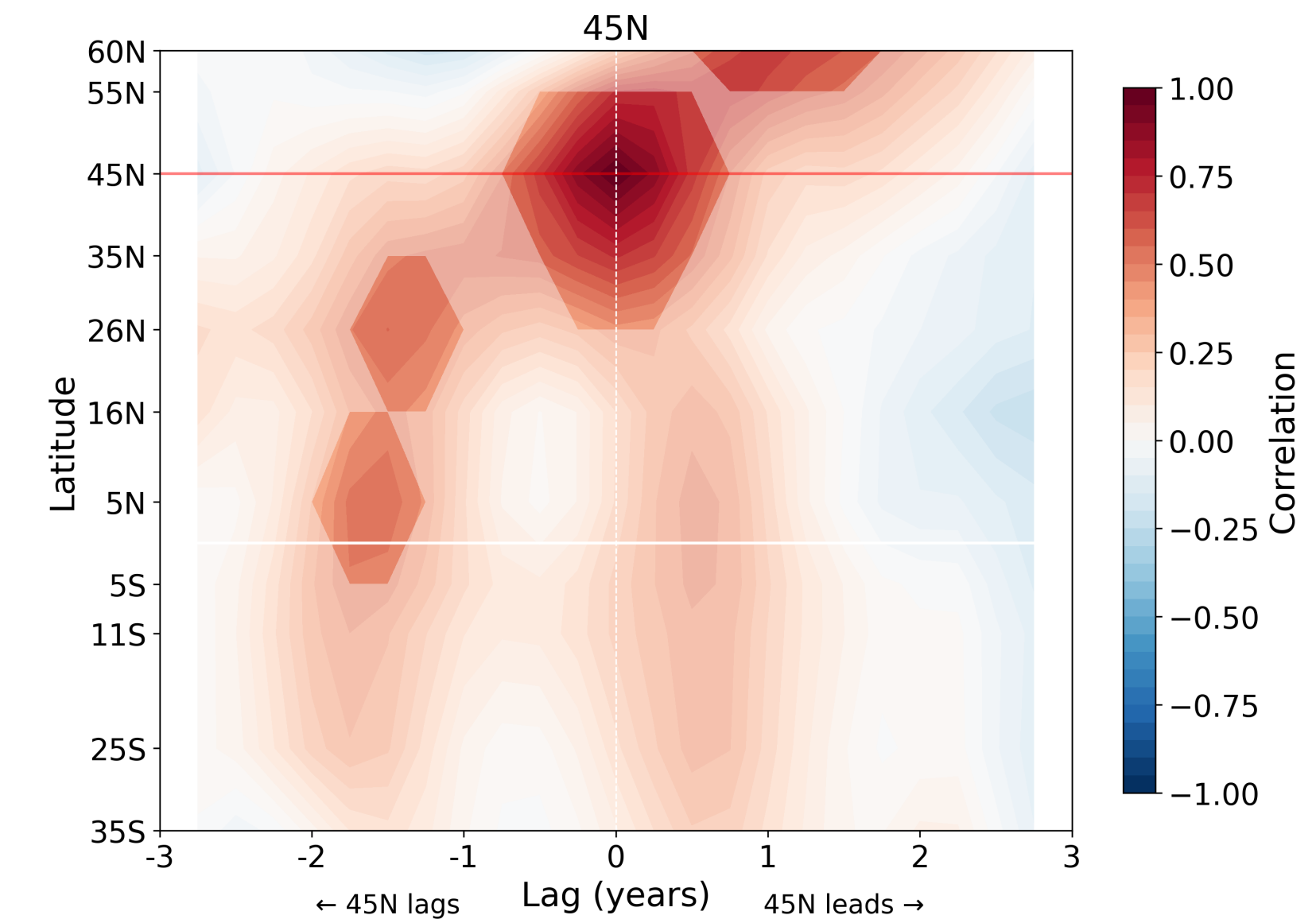
Comparing to Literature

Zhang (2010):

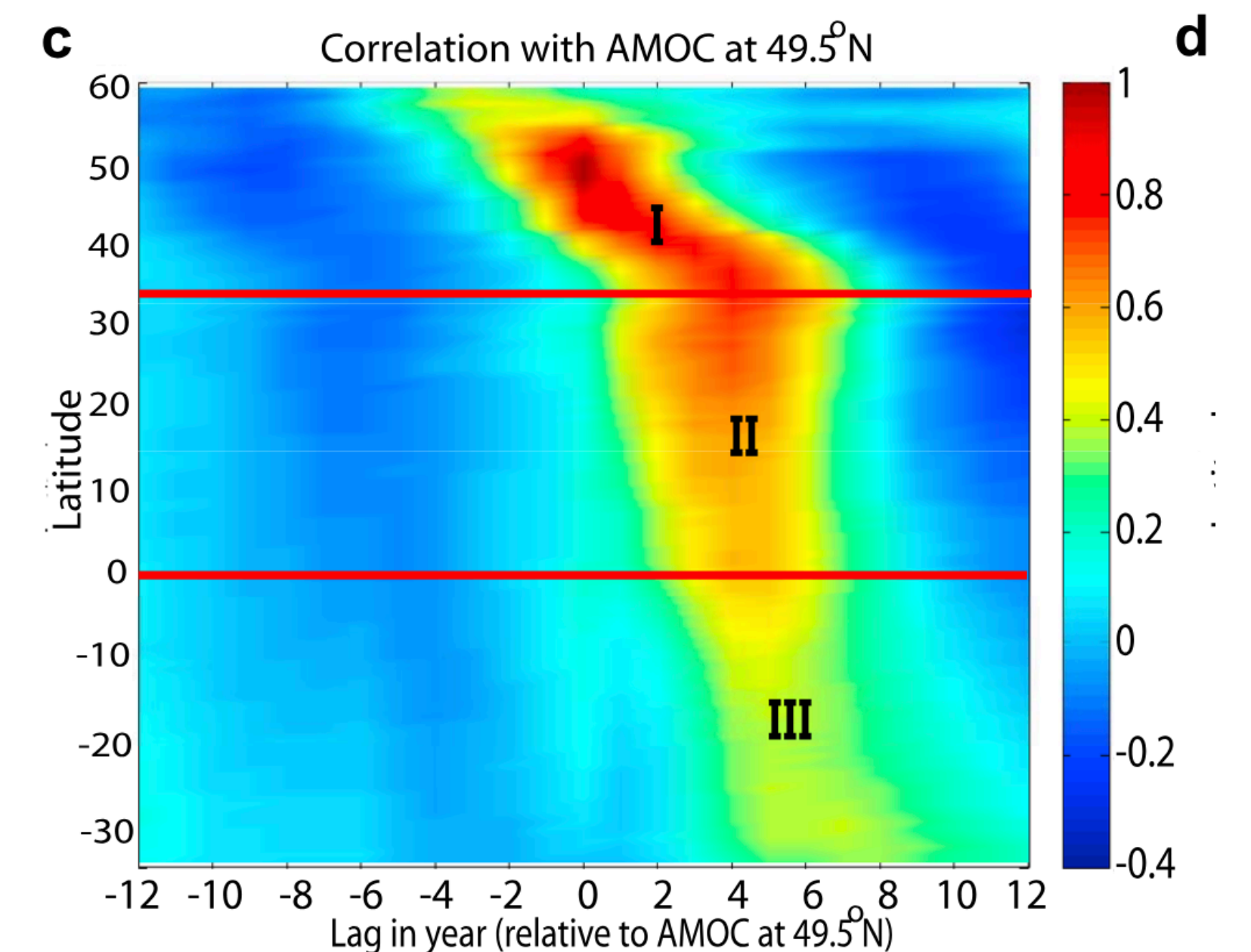
- Lead/lag patterns do **not match** results found from Zhang (2010) from AMOC model simulations, she finds:
 - „AMOC variations at 34°S lags those at the equator by about 3 years“
 - „... there is a 4-year lead between AMOC variations at 50°N and at 34°N.“
 - Northern latitudes leading those more southern, by much longer timescales (3-4 years)
- I find 45°N rather **lags** equatorward latitudes by 1.5 years, 35°S shows no lag but coherence with equatorward latitudes

Kelly et al. (2014):

- She found MHT anomaly coherence with southward source, until 40°N, then dropping - matches with my findings (so far)



Lead/lag plot using MHT from Calafat et al. (2025) for 45°N



Zhang, R. (2010), doi:10.1029/2010GL044474